Hourly Capital Bikeshare Demand Forecasting – Washington, DC

Abstract: Bike-sharing systems rely on accurate demand forecasts to maintain the availability of bicycles and docking spaces throughout the day. Short-term ridership fluctuates with weather conditions, commuter activity, and seasonal patterns, making it difficult for operators to manage supply without data-driven predictions. Prior studies show that temperature, humidity, rainfall, and time-of-day trends significantly affect hourly bike usage, emphasizing the importance of predictive modeling for operational planning [1], [2]. This study develops a supervised machine-learning approach to forecast hourly bike rentals in Washington, DC using the Capital Bikeshare dataset. The framework integrates temporal, calendar, weather, and lag-based features and is evaluated using a strictly chronological split to reflect real deployment conditions. The goal is to produce a practical and reliable 24-hour-ahead forecasting system that supports resource allocation and improves daily operational efficiency in urban bike-sharing networks.

1. Introduction

Bike-sharing programs have become a key component of urban mobility, providing flexible and affordable options for commuting, short-distance travel, and recreation. Their effectiveness depends on anticipating fluctuations in hourly demand so that bicycles and docking spaces remain available across the network. Prior research shows that environmental factors such as temperature, humidity, and rainfall, combined with routine commuting patterns embedded in daily and weekly cycles, strongly influence short-term ridership in shared-mobility systems [1]. When these variations are not anticipated, stations may become empty or congested at critical times, reducing service reliability and increasing operational strain.

Accurate demand forecasting helps operators prepare for upcoming periods of high or low usage, schedule bicycle rebalancing, and deploy staff more efficiently. Studies consistently emphasize that minimizing station imbalances and improving overall system availability require reliable predictions of near-future demand [2]. As bike-sharing systems grow, integrating weather information, calendar structure, and recent historical activity into machine-learning models has proven effective for capturing the complex relationships that drive hourly usage patterns.

This study addresses the problem of forecasting hourly bike rentals in Washington, DC, a mature bike-sharing network with rich historical, environmental, and temporal data. To meet this objective, a supervised machine-learning pipeline is developed that incorporates time-of-day and seasonal effects, weather observations, and lag-based features derived from past behavior. In addition to generating continuous 24-hour-ahead predictions, the system identifies high- and low-demand periods to support proactive operational decision-making. A strictly future test window is used to ensure that evaluation reflects real forecasting behavior. The broader aim is to develop a dependable and interpretable forecasting system that strengthens day-to-day operational planning within the Capital Bikeshare network.

2. Methodology

This study employs a supervised machine learning approach to forecast hourly bike-sharing demand in Washington, DC using the Capital Bikeshare dataset. The methodology follows a structured process that includes data collection, cleaning, exploratory analysis, feature engineering, model selection, and strictly time-ordered training and evaluation workflow. Each step is designed to preserve the temporal structure of the data, avoid information leakage from the future, and build a forecasting system that is realistic for operational deployment.

2.1. Dataset Description

The dataset used in this project is the Bike Sharing Dataset, which contains hourly rental records from the Capital Bikeshare system in Washington, DC. It includes 17,379 observations and 17 variables covering the period from January 2011 to December 2012. Each record provides information related to date and time (season, month, weekday, holiday, working day, and hour), along with environmental factors such as temperature, “feels-like” temperature, humidity, windspeed, and weather category.

The dataset also reports three usage counts - casual rentals, registered rentals, and total rentals (cnt), which serve as the target variable for forecasting. Because it spans two full years and captures different weather conditions and commuter patterns, the dataset is suitable for studying hourly demand behavior and developing machine-learning models for short-term forecasting [3].

2.2. Data Cleaning

The raw dataset was first inspected for structural issues. After loading the hour.csv file, duplicate rows were checked using drop_duplicates(). Since no duplicates were detected, the row count remained unchanged at 17,379. The next step involved fixing date–time information. The original date field dteday was converted into a valid datetime format, and a unified timestamp column, datetime, was created by adding the hour (hr) as a time offset. This combined timestamp provides a clear and consistent hourly index for all subsequent time-series processing.

To ensure correct variable types, all categorical or discrete columns such as season, yr, mnth, hr, holiday, weekday, workingday, and weathersit - were explicitly cast to integer types. Continuous variables, including temperature, humidity, windspeed, and rental counts, were converted to floating-point values. This step ensures that the data behaves consistently during modeling and avoids silent conversion errors. Rows with missing values in essential fields, specifically datetime or the target variable cnt, were removed; however, the dataset did not contain any such missing entries.

Additional cleaning steps focused on maintaining valid physical values and preparing the dataset for forecasting. Records with negative rental counts were removed, although none were present in the file. The dataset was then sorted chronologically using the datetime column to preserve the natural order required for time-dependent modeling. Outliers in hourly rental counts were not removed but were flagged using the interquartile range (IQR) method, resulting in 505 hours marked as unusually high or low demand. Keeping these observations is important because peak hours are genuine operational events.

A timeline completeness check revealed 165 missing hours between January 2011 and December 2012. These missing timestamps were only reported and not imputed, ensuring the dataset remains aligned with the original system records. Finally, ID-like or leakage-prone variables like instant, casual, and registered were removed to avoid unintended influence on the model. After all cleaning steps, the dataset retained 17,379 rows and 16 columns, covering the full operational period with a consistent, well-prepared time index suitable for machine-learning analysis.

2.3. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted to understand the structure of the cleaned dataset and identify key behavioral patterns in hourly bike rentals. Initial inspection of the first few rows and descriptive statistics confirmed that the dataset spans two full years with 17,379 hourly observations and includes temporal, weather, and demand-related variables. The summary statistics also highlighted the strong variability in bike demand, with rental counts ranging from 1 to 977 per hour.

2.3.1. Distribution of Hourly Rentals (cnt)

The histogram (fig1) of the target variable shows a right-skewed distribution, indicating that most hours have relatively low rental counts, while extremely high-demand periods occur less frequently. This skewness is typical for urban mobility data, where off-peak hours dominate the dataset and only specific commuting periods show sharp demand surges. The boxplot (fig 2) supports this pattern and visually confirms the presence of high-value observations, which were expected and therefore retained for modeling since they represent real peak-usage events.

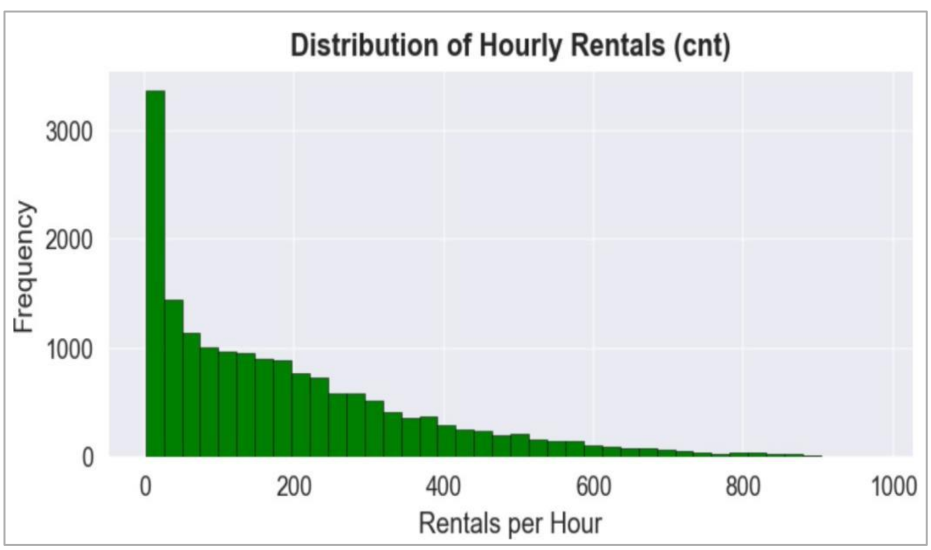

Fig 1 - Distribution of Hourly Rentals (cnt)

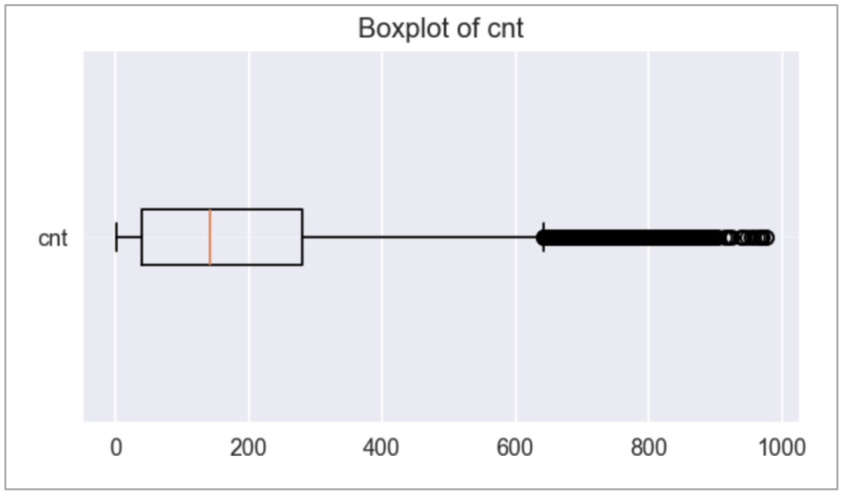

Fig 2 – Boxplot of cnt

2.3.2. Temporal Patterns: Hourly, Weekly, and Monthly Trends

Temporal aggregations reveal clear behavioral patterns in rider activity. Hourly averages (Fig 3) show two prominent peaks - a morning spike around 8-9 AM and a stronger evening spike around 5-6 PM, reflecting commuter rush hours. Weekly trends indicate higher rentals on weekdays, especially mid-week (Tuesday–Thursday), with slightly lower activity on weekends. Monthly averages show a strong seasonal pattern, with demand increasing steadily from early spring, peaking during summer months (June–August), and declining again during winter. These trends validate the importance of including time-based features in forecasting models.

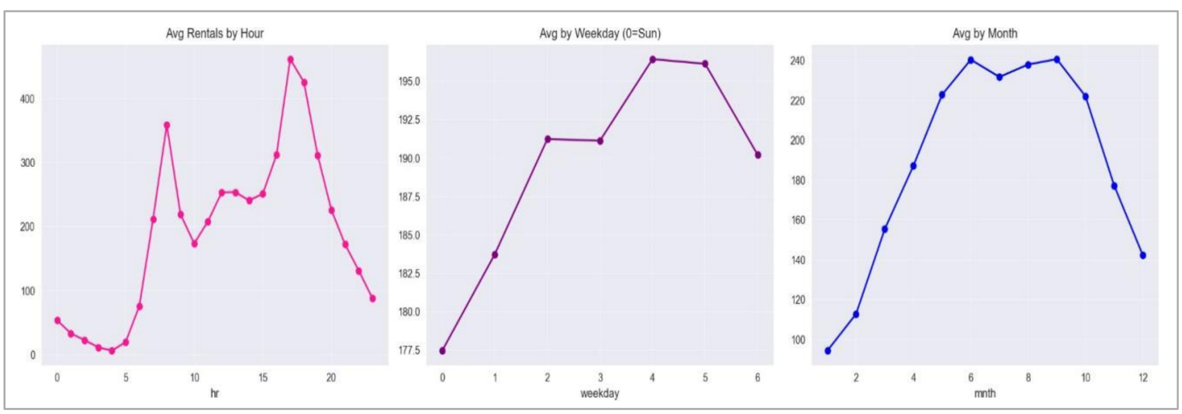

Fig 3 – Hourly, weekday, Monthly Patterns

2.3.3. Weather Influence on Rentals

Scatter plots (Fig 4) comparing rentals with weather variables demonstrate meaningful relationships. Rentals increase with temperature and feels like temperature, suggesting that warmer weather encourages outdoor travel. Conversely, high humidity and strong windspeed show a mild negative relationship with demand, as extreme weather tends to reduce ridership. These visual patterns justify the inclusion of weather-related predictors in the final model.

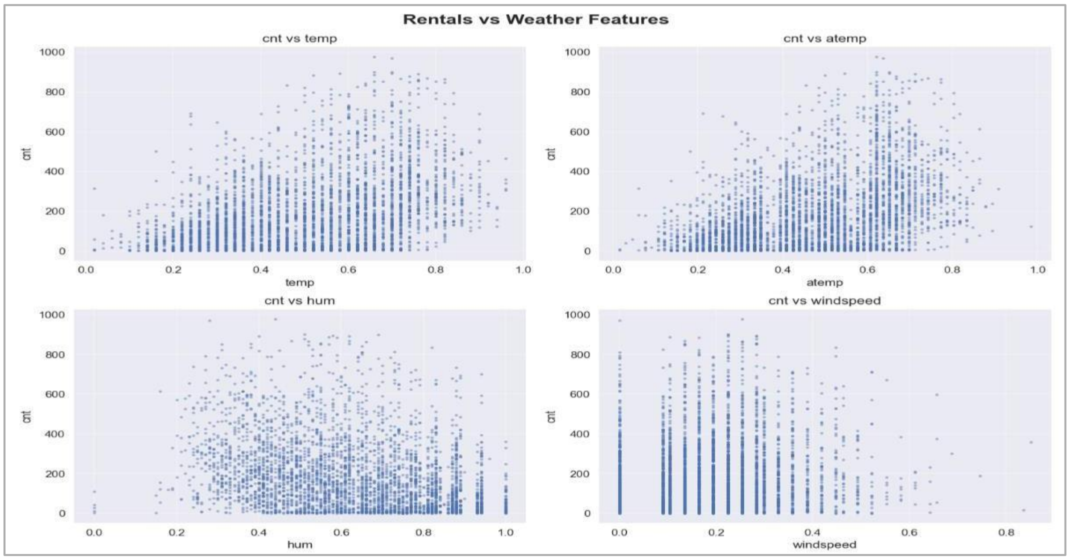

Fig 4 - Rentals vs Weather Variables

2.3.4. Correlation Analysis

The correlation heatmap (Fig 5) highlights several important relationships among numerical features. The strongest correlations appear between temp and atemp, as expected, while moderate positive correlations exist between temperature-related variables and cnt. Humidity and windspeed show weak or negative correlations with demand. Overall, the heatmap confirms that no severe multicollinearity exists among predictors, and that weather and time-based variables contribute distinct information to the forecasting problem.

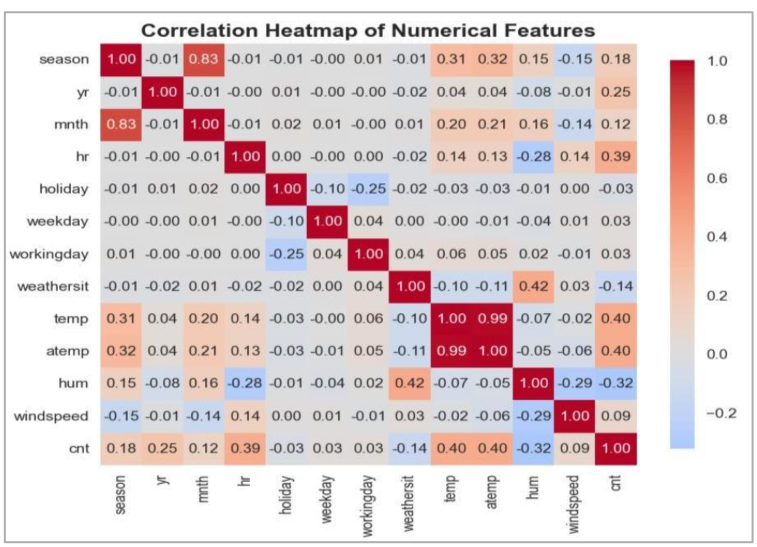

Fig 5 – Correlation Heatmap

2.3.5. Time-Series Trend Visualization

Year-wise line plots (Fig 6) for 2011 and 2012 display the evolution of hourly demand over time. Both years exhibit recurring daily demand cycles and seasonal fluctuations. The amplitude of peaks appears larger in 2012, indicating a general rise in system usage over time. These visual trends reinforce the importance of respecting temporal order during train-test splitting and support the use of rolling or lag-based features to capture historical dependencies

Fig 6 – Time Series Overview

2.4. Data Preparation

The data preparation stage focused on transforming the cleaned dataset into a structure suitable for time-series forecasting. Because future demand prediction requires respecting temporal order, the dataset was split chronologically. The final 50 days of data (November 11, 2012 to December 31, 2012) were reserved as an unseen test period, while all earlier observations were used for model training. This ensures that the model is evaluated on true future values rather than randomly shuffled data.

Feature engineering was applied to capture important temporal, behavioral, and environmental patterns that influence hourly bike rentals. First, cyclic encodings were generated for hour, day of week, and month to preserve the circular nature of time-based variables - such as hour 23 being close to hour 0. Additional calendar features were created, including indicators for federal holidays and weekends. A global trend variable (days_since_start) was added to represent long-term increases in bike-sharing usage over the two-year period.

Weather-related interaction features were also generated to help the model capture non-linear effects. For example, a derived “feels-like” temperature combines actual temperature and humidity, and polynomial temperature terms (temp_sq, temp_cubed) allow the model to represent curved relationships between temperature and demand.

To incorporate historical behavior, lag features were created at 1, 2, 24, 48, and 168 hours. These represent previous usage at immediate, daily, two-day, and weekly intervals. Rolling-window statistics were also included, such as 3-hour, 12-hour, and 24-hour moving averages, as well as 24-hour rolling standard deviation and maximum. All rolling features were shifted by one hour to ensure that the model never uses future information.

Categorical variables such as season, weather category, month, and weekday were one-hot encoded to convert them into machine-readable form without imposing false numeric relationships. Hour was not one-hot encoded to avoid unnecessary dimensionality since its cyclic encoding already captures temporal patterns.

After feature construction, the full dataset was split into training and testing subsets based on the temporal cutoff. Rows containing NaN values - created by lag and rolling operations were removed to avoid leaking partial history into the model. The final feature matrix consisted of 55 engineered predictors. The training set contained 16,013 rows, while the test set contained 1,198 rows representing the unseen last 50 days.

A history of past cnt values from the training set was also stored to support live forecasting during model evaluation.

2.5. Model Selection

In this project, a diverse set of six supervised regression models was selected to represent different learning strategies and levels of model complexity. The objective was to compare the performance of simple linear techniques with more flexible tree-based and gradient-boosting methods for forecasting hourly bike-sharing demand. Using a varied set of algorithms allows the study to assess not only which model performs best, but also how different modeling approaches handle the temporal, weather-based, and engineered features present in the dataset.

Ridge Regression was included as a reliable linear baseline, offering a reference point against which the gains from more complex models could be measured. Random Forest was chosen for its ability to capture non-linear interactions by aggregating predictions from many deep decision trees, making it robust to noise and overfitting. Histogram Gradient Boosting was added as an efficient implementation of gradient boosting, particularly suitable for continuous numerical features. XGBoost and LightGBM were selected due to their strong empirical performance on large, structured datasets and their ability to model subtle relationships through iterative boosting. Finally, CatBoost was incorporated because of its consistent success in tabular, time-related prediction problems and its ability to model non-linear patterns with limited pre-processing.

Evaluating this balanced set of linear, bagging, and boosting models ensures a comprehensive and fair comparison across multiple algorithmic families, providing a solid foundation for selecting the most reliable forecasting method for hourly bike-sharing demand [4].

2.6. Model Building & Training

To establish a fair comparison across algorithms, we first built baseline models using a common feature set, a fixed train–test split (training on the historical period and testing on the last 50 days), and the same target transformation. For each model in the baseline_models dictionary (Ridge, Random Forest, HistGradientBoosting, XGBoost, LightGBM, and CatBoost), the model was fitted on X train with the transformed target Log (1 + cnt). Predictions on X test were then mapped back to the original rental scale with the inverse transformation exp(.) -1

Within a single training loop, the code recorded four metrics for every model: symmetric mean absolute percentage error (sMAPE, %), mean absolute error (MAE), root mean squared error (RMSE), and training time (seconds). These results (Fig 7) were stored in a list, converted into a DataFrame, and sorted by sMAPE to produce baseline leaderboard. The baseline results show that the linear Ridge model performs poorly on this non-linear problem (test sMAPE ≈ 49.2%, RMSE ≈ 135.6), while all tree-based and boosting models achieve test sMAPE values around 20 - 21%. Among the baselines, CatBoost is the strongest model, with a test sMAPE of 19.752%, MAE ≈ 21.76, and RMSE ≈ 34.85, with the longest training time (~210 s). These baseline scores provide a reference point for judging whether hyperparameter tuning meaningfully improves performance.

Baseline Leaderboard (lower sMAPE is better):

	Model	sMAPE (%)	MAE	RMSE	Time (s)
0	CatBoost	19.752	21.76	34.85	210.2
1	XGBoost	20.177	21.66	34.85	42.8
2	Random Forest	20.317	24.12	40.46	43.3
3	HistGB	20.506	23.69	37.16	9.4
4	LightGBM	20.848	22.89	35.97	73.3
5	Ridge	49.225	76.57	135.56	0.0

Fig 7 – Baseline Results

2.6.1 Model Iteration & Hyperparameter Tuning

After obtaining the baseline performance for all six models, the next step was to improve them through targeted hyperparameter optimization. Each algorithm was tuned using Optuna, which automatically searches for parameter combinations that minimize validation error. To ensure that tuning reflects real forecasting behavior, the workflow used a 5-fold TimeSeriesSplit, where every fold trains on earlier timestamps and validates on strictly later observations. This structure preserves the direction of time and prevents data leakage that would artificially inflate accuracy.

For each trial in the search process, Optuna proposes a set of hyperparameters, the model is trained on the training portion of each fold, and predictions are evaluated using sMAPE on the validation segment. The mean sMAPE across all five folds becomes the objective value for that trial. The number of search trials matches the complexity of each model: Ridge Regression was tuned for 20 trials, while the non-linear models - Random Forest, HistGradientBoosting, XGBoost, LightGBM, and CatBoost were tuned for 40 to 50 trials. Once the study identifies the best configuration, the corresponding model is retrained on the entire training period and evaluated on the unseen 50-day test window.

The tuned results show (Fig 8) that hyperparameter optimization yields meaningful improvements for all non-linear models. While Ridge Regression remains substantially less accurate, the boosting-based models converge around test sMAPE values near 20%, with CatBoost achieving the strongest performance at 19.222%. This consistent improvement across boosted learners demonstrates the value of tuning in capturing the complex interactions present in hourly bike-sharing demand.

FINAL HYPERPARAMETER-TUNED LEADERBOARD (Test sMAPE)

	Model	sMAPE (%)	CV sMAPE (%)	Time (s)	Best Params
0	CatBoost	19.222	20.136	24.4	{'iterations': 2675, 'learning_rate': 0.013378765218633711, 'depth': 6, 'l2_leaf_reg': 9.808883849686215, 'border_count': 32}
1	Random Forest	19.898	20.998	50.3	{'n_estimators': 961, 'max_depth': 35, 'min_samples_split': 4, 'min_samples_leaf': 3, 'max_features': 0.8}
2	HistGB	20.300	20.199	20.5	{'max_iter': 2465, 'learning_rate': 0.010784820891833686, 'max_depth': 22, 'l2_regularization': 0.3261879882725953}
3	XGBoost	20.359	19.859	13.8	{'n_estimators': 2762, 'learning_rate': 0.008117298797571296, 'max_depth': 7, 'subsample': 0.9198001771415915, 'colsample_bytree': 0.8156213181164734, 'gamma': 0.03509098942566083, 'min_child_weight': 10}
4	LightGBM	20.443	19.789	4.7	{'n_estimators': 2119, 'learning_rate': 0.00595263181050145, 'num_leaves': 235, 'max_depth': 6, 'min_child_samples': 37, 'subsample': 0.6697040308295864, 'colsample_bytree': 0.6248860498519436}
5	Ridge	49.219	54.043	0.0	{'alpha': 0.10484580986876234}

Fig 8 – Tuned Model Results

3. Results and Visualization

This section presents the final performance of all tuned models on the 50-day unseen test set and summarizes how the selected model behaves under realistic forecasting scenarios. All results reflect true out-of-sample performance because the test set was never used during training, tuning, or feature engineering.

3.1. Best Regression Model

The final evaluation on the 50-day unseen test set identified CatBoost (Fig 9) as the best-performing regression model for forecasting hourly bike rentals. After hyperparameter tuning, CatBoost achieved a test sMAPE of 19.22%, which closely matches its cross-validated sMAPE of 20.14%. This small difference indicates that the model generalizes well and does not overfit the training data. The model also reached a directional accuracy of 80.78%, showing that it reliably predicts both the magnitude of demand and the direction of hour-to-hour changes. Based on this consistent performance across multiple evaluation measures, CatBoost was selected as the final regression model for all downstream forecasting analyses in this study.

```
Best MODEL → CatBoost
   Test sMAPE: 19.222% | CV sMAPE: 20.136%
   Approx. accuracy: 80.778%
```

Fig 9 – Best Model

3.2. Demand-Level Classification Accuracy

For operational use, continuous predictions were grouped (Fig 10) into demand categories using training-set quartile thresholds, hours above the 75th percentile were labeled HIGH, those below the 25th percentile were labeled LOW, and the rest were marked as NORMAL. This allows the model's output to be interpreted in terms of typical low, regular, and high demand periods.

When CatBoost predictions were compared with the true categories in the unseen 50-day test set, the model reached an accuracy of 91.7%. Most errors occurred within the NORMAL range, while incorrect assignments between LOW and HIGH were very rare. This indicates that the model reliably distinguishes quiet hours from peak-demand periods.

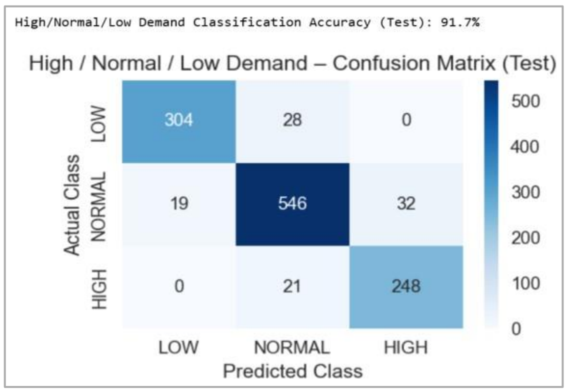

Fig 10 - Classification Accuracy

3.3. Weather-Sensitivity Analysis

A focused weather-sensitivity test (Fig 11) was performed to examine how predicted demand changes with temperature during the 8 AM rush hour. Holding all other features constant at median test-set values, temperature was varied from 0°C to 35°C. Predictions ranged from approximately 493-498 rentals at colder temperatures, increasing to a baseline of 517 rentals at 15°C (59°F) and peaking near 524 rentals demand gradually decreases again at 20°C (68°F). At higher temperatures, predicted, falling to approximately 494 rentals at 35°C. This pattern indicates that moderate temperatures between 15-20°C are most favorable for morning ridership, whereas both extreme cold and heat slightly suppress demand.

WEATHER SENSITIVITY ANALYSIS – 8 AM RUSH HOUR

	Temperature (°F)	Temperature (°C)	Predicted 8 AM Demand	Change vs 59°F
0	32°F	0°C	493 bikes	—
1	41°F	5°C	492 bikes	—
2	50°F	10°C	498 bikes	—
3	59°F	15°C	517 bikes	Baseline
4	68°F	20°C	524 bikes	+7 bikes
5	77°F	25°C	514 bikes	-3 bikes
6	86°F	30°C	508 bikes	-9 bikes
7	95°F	35°C	494 bikes	-23 bikes

Fig 11 - Weather-Sensitivity Analysis

3.4. Sample 24-Hour Forecast Using Live Weather Data

To demonstrate real-time deployment capability, the final CatBoost model was combined with live weather inputs retrieved through the Open-Meteo API. In this setup, the historical dataset is used exclusively for model training, while the forecast itself begins from the current real-world date and time. The next 24 hourly timestamps are generated automatically, and each hour is paired with live temperature, humidity, windspeed, and weather descriptions from the API. These enriched features are then passed through the trained model to produce a forward-looking 24-hour rental forecast for Washington, DC.

The resulting time-series output (Fig. 12) reflects realistic short-term demand behavior. The forecast shows a moderate evening peak on the first day, followed by a gradual decline into the early morning hours where predicted usage remains low. Around 6–7 AM, demand begins to rise sharply, leading to a clear morning peak between 8–9 AM with predicted values in the range of 180–200 rentals per hour. This pattern aligns well with the weekday rush-hour dynamics captured in the historical dataset. The accompanying table (Fig. 13) lists each hourly prediction alongside temperature and simplified weather categories, offering contextual information that can support short-term operational decision-making.

However, since the historical dataset used for model training and validation ends in 2012 and no real-time ground-truth rental data available for future dates, these live forecasts cannot be directly validated. However, the model's reliability has already been demonstrated on the 50-day test window within the historical dataset, where it closely tracked real hourly demand trends and achieved approximately 80% accuracy, as shown in the Actual vs. Predicted plot (Fig. 15). This evaluation indicates that the model can generate meaningful short-term forecasts when supplied with real-time weather information.

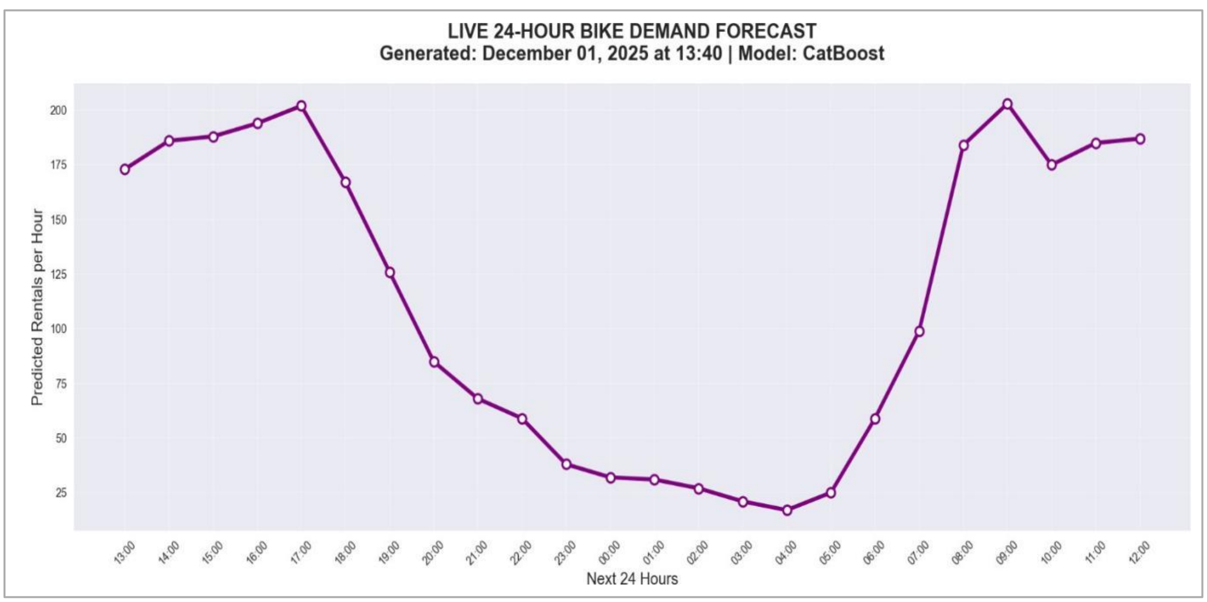

Fig 12 - Live 24- hour Forecast

NEXT 24 HOURS — HOURLY PREDICTION

LIVE 24-HOUR DEMAND FORECAST

	Time	Predicted Demand	Temperature (°F)	Weather
0	Dec 01, 13:00	173	44.8°F	Unknown
1	Dec 01, 14:00	186	45.3°F	Unknown
2	Dec 01, 15:00	188	45.3°F	Unknown
3	Dec 01, 16:00	194	44.4°F	Unknown
4	Dec 01, 17:00	202	40.8°F	Unknown
5	Dec 01, 18:00	167	38.7°F	Unknown
6	Dec 01, 19:00	126	36.3°F	Unknown
7	Dec 01, 20:00	85	34.9°F	Unknown
8	Dec 01, 21:00	68	33.8°F	Unknown
9	Dec 01, 22:00	59	33.1°F	Unknown
10	Dec 01, 23:00	38	32.4°F	Unknown
11	Dec 02, 00:00	32	32.2°F	Unknown
12	Dec 02, 01:00	31	31.8°F	Unknown
13	Dec 02, 02:00	27	31.8°F	Unknown
14	Dec 02, 03:00	21	31.5°F	Unknown
15	Dec 02, 04:00	17	31.6°F	Unknown
16	Dec 02, 05:00	25	31.1°F	Unknown
17	Dec 02, 06:00	59	32.7°F	Unknown
18	Dec 02, 07:00	99	34.0°F	Unknown
19	Dec 02, 08:00	184	34.7°F	Unknown
20	Dec 02, 09:00	203	36.0°F	Unknown
21	Dec 02, 10:00	175	36.9°F	Unknown
22	Dec 02, 11:00	185	37.8°F	Unknown
23	Dec 02, 12:00	187	39.2°F	Unknown

Fig 13 - table lists

3.5. Model Performance Comparison

The tuned model comparison (Fig 14) summarizes the test sMAPE values for all six algorithms - Ridge, Random Forest, Histogram Gradient Boosting, XGBoost, LightGBM, and CatBoost. Among these, CatBoost achieves the lowest error, with a test sMAPE near 19.2%, outperforming the other tree-based methods, which cluster around the 20-20.5% range. Ridge Regression performs substantially worse (around 49% sMAPE), demonstrating that simple linear assumptions do not sufficiently capture the nonlinear and interacting effects present in the dataset. This comparison confirms that boosting-based nonlinear models are more suitable for this forecasting problem.

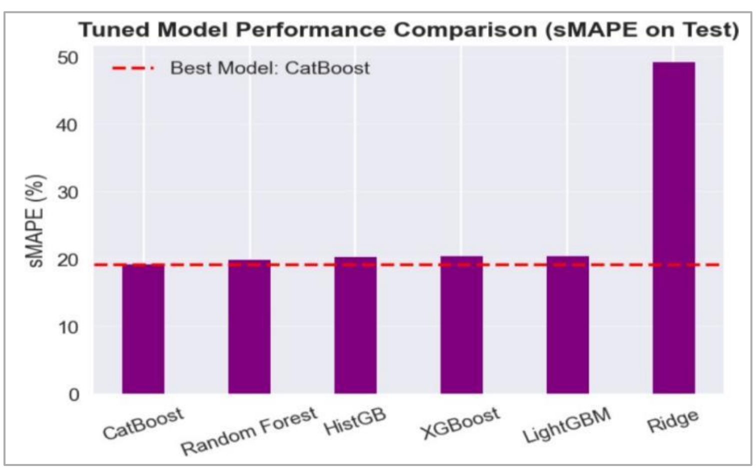

Fig 14 – Model Comparison

3.6 Actual vs Predicted Rentals

To visually assess the temporal behavior of the final model, Fig 15 presents actual and predicted hourly rentals across the entire 50-day test window. The figure is divided into two panels: the first half of the test period and the second half. In both panels, the model successfully tracks daily rental cycles, including morning peaks, evening peaks, and nighttime lows. Although occasional deviations occur, there is no systematic drift or bias, and the model remains closely aligned with actual demand patterns. This visual consistency supports the numerical sMAPE result and demonstrates that CatBoost captures underlying temporal structure effectively.

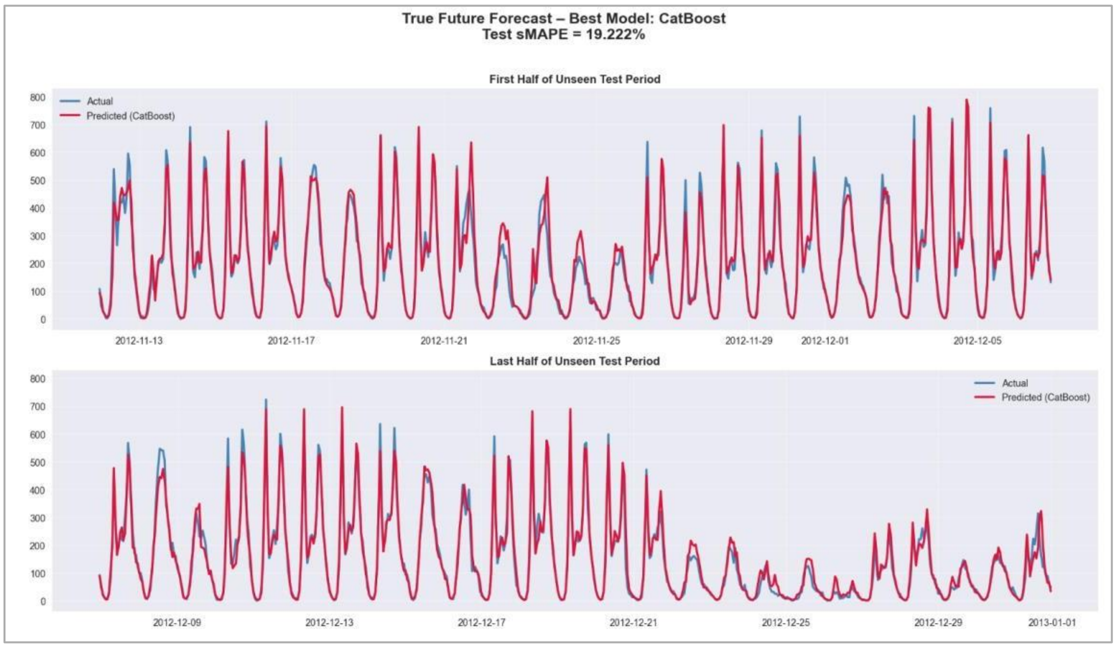

Fig 15 – Actual vs Predicted Rentals

4. Discussion & Contribution

The forecasting results indicate that the tuned CatBoost model provides the most accurate and stable predictions among all evaluated models. Its low sMAPE on the unseen 50-day test set, along with the close alignment between actual and predicted curves, confirms that the model successfully captures the hourly structure of bike-sharing demand. The model reproduces key temporal dynamics - morning and evening peaks, lower night usage, and shifts across weekdays - without consistent drift or instability. Additional analyses, such as the demand-level classification and temperature-sensitivity study, further reinforce the model's ability to translate numeric predictions into meaningful operational insights.

The primary goal of this project was to build a reliable short-term (24-hour ahead) forecasting system for hourly bike-sharing demand. The results demonstrate that this objective was achieved: the model generalizes well to future conditions, performs strongly on a strictly chronological test window, and maintains high accuracy when converted into LOW, NORMAL, and HIGH demand categories. The sample 24-hour forecast generated using live weather data also shows that the model can be integrated into a practical decision-support workflow. Overall, the problem was effectively addressed within the scope of the available dataset.

This project contributes a complete and rigorously evaluated forecasting framework tailored for hourly bike-sharing demand. The work integrates several methodological elements that extend beyond simple regression pipelines - a feature-engineering design that combines cyclic encodings, weather interactions, calendar effects, lag features, and rolling statistics; a fully chronological evaluation scheme that avoids data leakage and more closely reflects real deployment; an operational demand-band classification layer that converts predictions into actionable categories; and a temperature-sensitivity analysis that helps interpret how ridership responds to weather changes. These components together provide a structured approach that can be adapted for demand forecasting in similar mobility systems.

Future enhancements can focus on expanding the dataset to include rental records beyond 2012, enabling evaluation under more recent behavioral and environmental conditions. Incorporating real-world ground-truth data for the generated 24-hour forecasts would allow continuous monitoring of accuracy under live conditions. Additional improvements may include integrating holiday/event calendars, exploring deep learning methods such as LSTMs or Temporal Convolutional Networks, and developing station-level or zone-level predictions for rebalancing. A broader extension could compare demand across multiple bike-sharing companies or cities, offering insight into competitive dynamics and regional variability in usage patterns.

5. Conclusion

This study demonstrates that a well-designed machine-learning pipeline can accurately forecast hourly bike-sharing demand and support operational planning. Among the evaluated models, CatBoost delivered the strongest performance on the unseen 50-day test window and consistently aligned with real usage patterns. The model captured essential temporal dynamics, provided reliable demand-band classification, and produced realistic 24-hour forecasts when paired with live weather data. While the historical dataset limits real-time validation, the results confirm that the approach is effective for short-term demand prediction and provides a solid foundation for future extensions, including live monitoring, expanded datasets, and more granular station-level forecasting.

Reference:

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